

5G/6G Communication Systems- ECA5601 (LAB)

Student Name: A.SAI KIRAN

Reg no: 192312672L

Experiment 10: 5G Network Slicing Environment Using MATLAB

Aim: To analyze the performance of a 5G Network Slicing environment by studying the number of network slices, slice utilization, and end-to-end latency using MATLAB.

Objective 1:

Program:

```
clc;

clear;

close all;

slices = {'eMBB','URLLC','mMTC','Private','IoT'};

count = [1 1 1 1 1]; % One instance of each network slice

figure

bar(count)

grid on

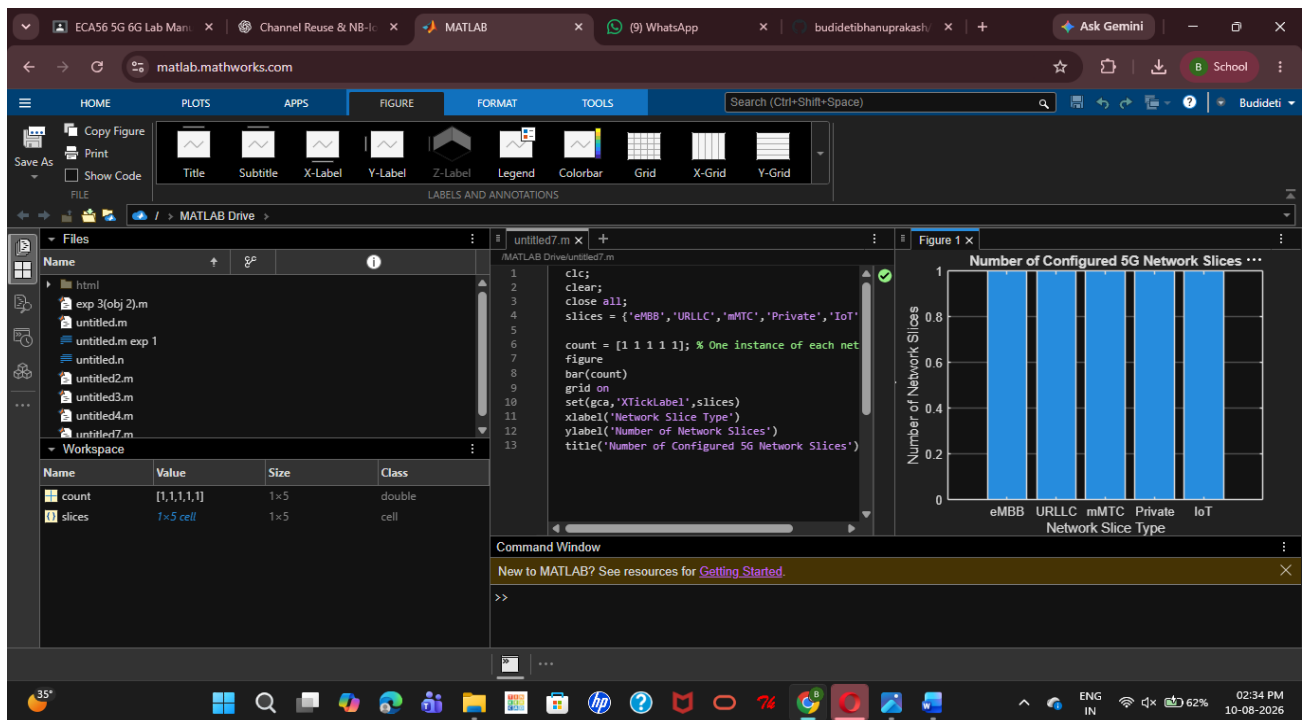
set(gca,'XTickLabel',slices)

xlabel('Network Slice Type')

ylabel('Number of Network Slices')

title('Number of Configured 5G Network Slices')
```

Output:



Objective 2:

Program:

```
clc;
```

```
clear;
```

```
close all;
```

```
slices = categorical({'eMBB','URLLC','mMTC','Private','IoT'});
```

```
utilization = [85 70 60 75 50];
```

```
figure;
```

```
bar(slices, utilization);
```

```
grid on;
```

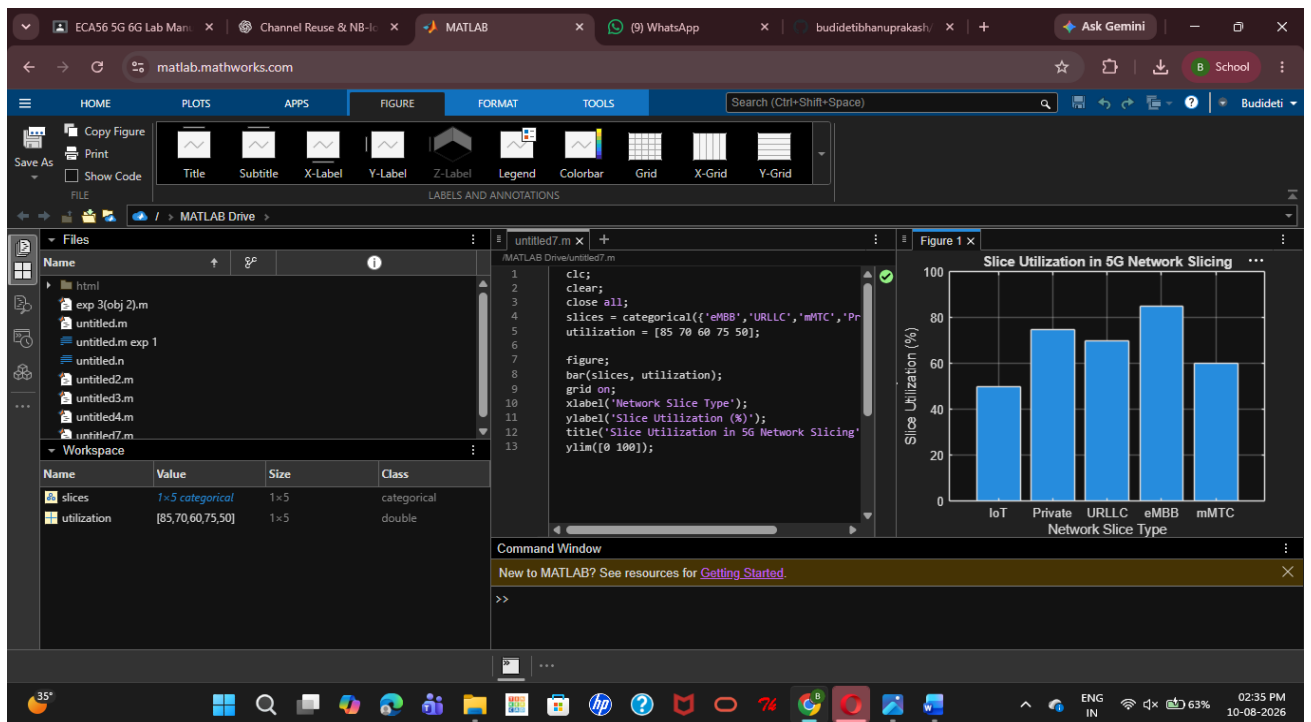
```
xlabel('Network Slice Type');
```

```
ylabel('Slice Utilization (%)');
```

```
title('Slice Utilization in 5G Network Slicing');
```

```
ylim([0 100]);
```

Output:



Objective 3:

MATLAB Code:

```
clc;

clear;

close all;

slices = [1 2 3 4 5];

latency = [25 5 40 15 60];

figure;

plot(slices, latency, '-o','LineWidth',2,'MarkerSize',8);

grid on;

xticks(slices);

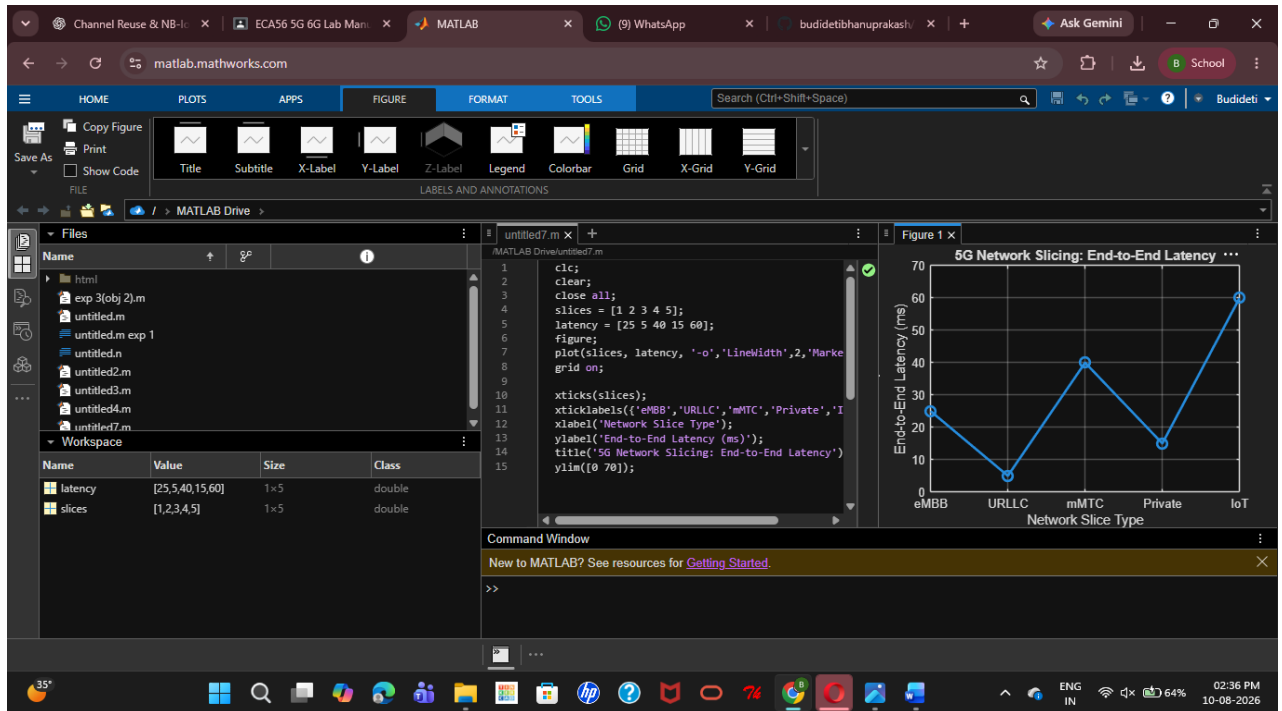
xticklabels({'eMBB','URLLC','mMTC','Private','IoT'});

xlabel('Network Slice Type');

ylabel('End-to-End Latency (ms)');
```

```
title('5G Network Slicing: End-to-End Latency');  
ylim([0 70]);
```

Output:



Result:

The 5G network slicing environment was successfully analysed in MATLAB, showing that multiple slices can be efficiently allocated, resource utilization can be compared, and end-to-end latency varies according to the QoS requirements of different 5G services.